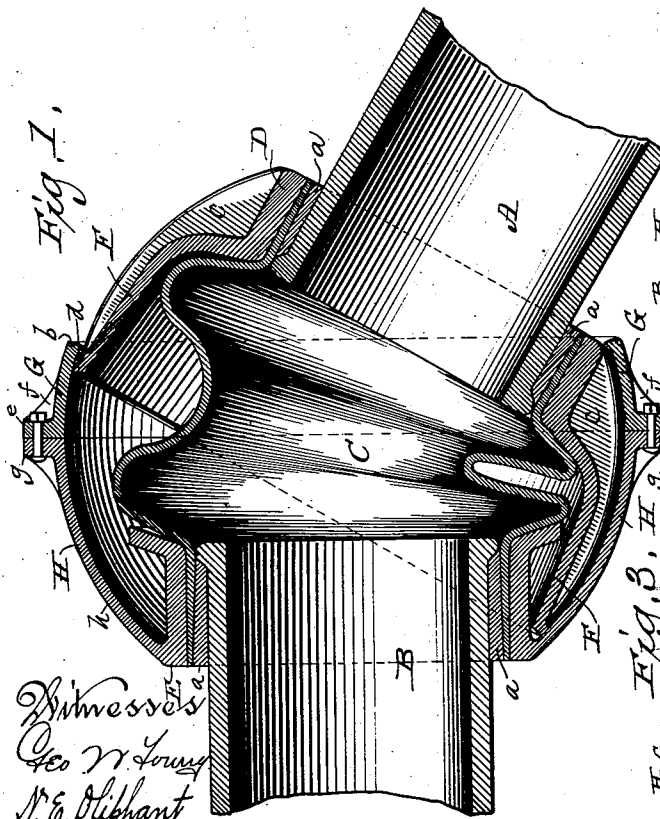
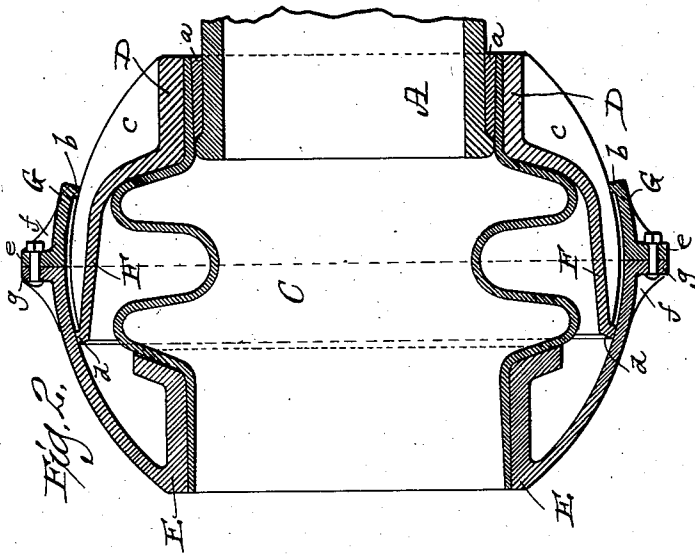


(No Model.)

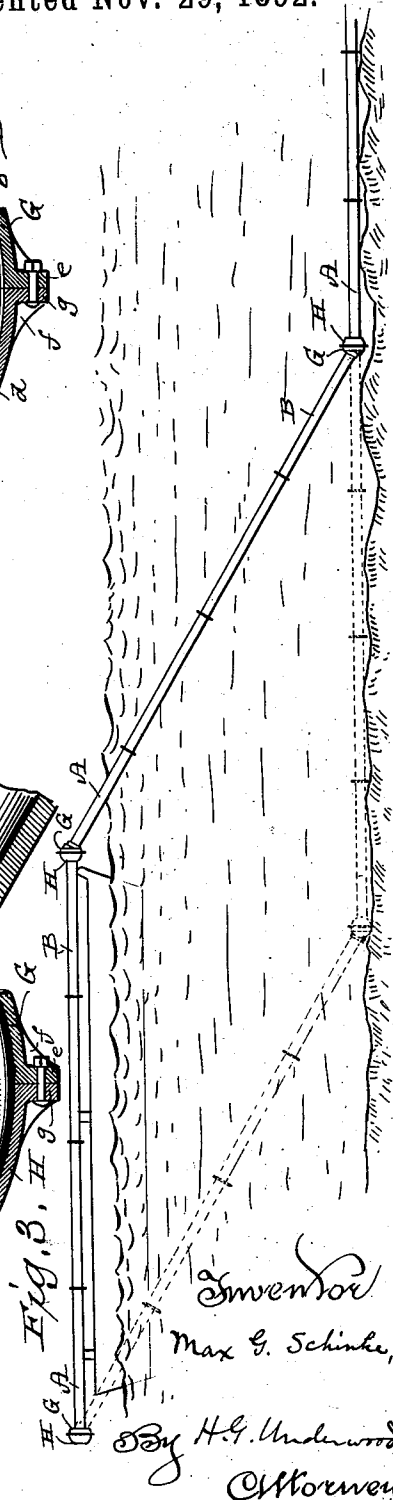
M. G. SCHINKE.
SUBMERGED PIPE.

No. 486,986.

Patented Nov. 29, 1892.



Witnesses
Geo. W. Young
N. E. Delphand



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UNITED STATES PATENT OFFICE.

MAX G. SCHINKE, OF MILWAUKEE, WISCONSIN.

SUBMERGED PIPE.

SPECIFICATION forming part of Letters Patent No. 486,986, dated November 29, 1892.

Application filed July 27, 1892. Serial No. 441,376. (No model.)

To all whom it may concern:

Be it known that I, MAX G. SCHINKE, a citizen of the United States, and a resident of Milwaukee, in the county of Milwaukee, and in the State of Wisconsin, have invented certain new and useful Improvements in Submerged Pipes; and I do hereby declare that the following is a full, clear, and exact description thereof.

My invention relates particularly to the laying of submerged pipe where heretofore the services of a diver have been required to adjust the pipe in position and calk the joints under water; and the object of said invention is to provide flexible water-tight joints (at intervals of the submerged-pipe system) having protective casings, the joints and casings being completed above water and then lowered to position.

Having stated the object of my invention, the latter consists in certain peculiarities of construction and combination of parts hereinafter described with reference to the accompanying drawings and subsequently claimed.

In the drawings, Figure 1 represents a sectional view of a flexible pipe-joint constructed according to my invention as it appears when two non-flexible sections of the joint are at an angle to each other; Fig. 2, a similar view of the joint on a straight line, and Fig. 3 a diagram illustrating my improved manner of laying submerged pipe.

Referring by letter to the drawings, A B represent opposing pipe-sections of non-flexible material—such as iron—that in practice are joined to an intermediate pipe-section C, of lead or other suitable flexible material, this latter pipe-section being preferably corrugated. As shown, the flexible pipe-section is joined at its ends to the adjacent ends of the opposing non-flexible pipe-sections by clamping-rings D E of suitable material—such as iron—the joints being calked with lead or other suitable material *a* in order to be water-tight, and, as is also shown, the clamping-rings are preferably integral with sections of spherical casings designed to protect said flexible pipe-section from external mechanical influences and forces, as well as to guide and limit the movement of the flexible water-

tight joint when being lowered to its submerged position, as will be hereinafter more fully described.

As illustrated, the ring D, that clamps the flexible pipe-section C to the non-flexible pipe-section A, has a rearward bell-like extension F, terminating in an outturned annular flange *d*, and external longitudinal ribs *c* are cast with said ring and its extension to form guides for an inturned annular flange *b* on a collar G, that is provided with an exterior flange *e*, strengthened by webs *f*, having a series of bolt-openings therein, these openings being designed for registration with similar openings in an opposing flange *g* on a bell-like forward extension H of the ring E, that clamps said flexible pipe-section to the non-flexible pipe-section B, the inner end of this latter ring being preferably in the form of a flange *h* in opposition to the corrugated portion of the aforesaid flexible pipe-section. The bell-like extensions of the clamping-rings form the spherical casing for the guidance of the flexible pipe-section, as well as its protection from external mechanical influences and forces, these sections of the casing having a sliding motion one upon the other to permit of the non-flexible pipe-sections being brought to various angles of deflection.

It is preferable to have the interior of the clamping-rings somewhat conical and to flange the ends of the non-flexible pipe-sections, these flanges serving as stops for the calking, while the conical interiors of said clamping-rings is a matter of detail and expediency toward obtaining the tightest possible joint between said non-flexible pipe-sections and the intermediate flexible one, the joint of the spherical protective casing being completed by bolting the flanged collar G to the flanged extension H of the clamping-ring E, above specified.

The work of laying the pipe being done from a staging or vessel, any convenient or desirable number of non-flexible pipe-sections may be joined together by other than flexible connections if preferred; but at certain intervals in the line of the pipe system I necessarily employ the flexible joint hereinbefore specified.

As shown in Fig. 3, a length of submerged pipe is lifted above water at one end (the other end of this length being also provided with a flexible joint) and held at an angle to the water-level. The present upper outer flexible pipe-section having been primarily prepared for the required angle of deflection and the new length brought to position to be joined, the several parts constituting the flexible joint and protective casing are put together in proper succession and said flexible pipe-section connected water-tight to the ends of the rigid sections. Upon completion of the flexible joint and its casing the same are lowered and the free end of the added length of non-flexible pipe held above water for immediate further connection or temporarily lowered to be lifted at a convenient time.

I am aware of various flexible pipe-couplings; but none of these are adapted to my purpose—the laying of submerged water or gas pipe of large size—and at this point in the description it is proper to emphasize the fact that the flexible pipe-section herein described is cast or rolled without joints. It is also to be observed that the protective casing is devoid of pivots and that the ends of the non-flexible pipe-sections do not have to be especially prepared in casting or otherwise for their connection with said flexible pipe-section, it being my purpose to flexibly join

sections of water or gas pipe such as is in common use.

Having now fully described my invention, what I claim as new, and desire to secure by Letters Patent, is—

1. In submerged pipe, the combination of two non-flexible pipe-sections, an intermediate flexible pipe-section engaging the ends of the former sections, but free to be bent to various angles, clamping-rings arranged on the engaged pipe-sections, and calking material interposed between the ends of said non-flexible and flexible pipe-sections, substantially as set forth.

2. In submerged pipe, two non-flexible pipe-sections, an intermediate flexible pipe-section engaging the ends of the former sections, but free to be bent to various angles, and suitable means for forming water-tight connection between said pipe-sections, in combination with a guiding and protective casing that incloses the joint formed by said pipe-sections, substantially as set forth.

In testimony that I claim the foregoing I have hereunto set my hand, at Milwaukee, in the county of Milwaukee and State of Wisconsin, in the presence of two witnesses.

MAX G. SCHINKE.

Witnesses:

N. E. OLIPHANT,
JOHN E. WILES.